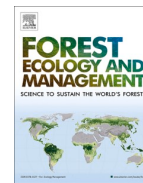




Contents lists available at ScienceDirect

Forest Ecology and Management

journal homepage: www.elsevier.com/locate/foreco

Allocation to tree bark in pine and oak species in fire affected mixed forests across the Northern Hemisphere

Ripu Daman Singh^a, Surabhi Gumber^{a,*}, Harshita Joshi^b, Surendra P. Singh^c^a Research Scholar, Department of Forestry and Environmental Science, DSB Campus, Kumaun University, Nainital 263001, India^b Research Scholar, Department of Botany, IPGGPG College of Commerce, Haldwani, Kumaun University, Nainital 263001, India^c INSA, Senior Scientist, Central Himalayan Environment Association (CHEA), Dehradun 248006, India

ARTICLE INFO

Keywords:

Adaptation to Forest fire
Northern Hemisphere
Oak-Pine forests
Relative Bark thickness

ABSTRACT

Pines (*Pinus*, with 120 species) and oaks (*Quercus*, with at least 400 species), the two principal forest forming genera of the Northern Hemisphere occur in fire affected landscapes over a large geographical area from 120°W to 150°E long, and from 10° to 60°N lat. We compiled studies of 116 such sites of which 75 sites had data of bark thickness. Taking a meta-analysis approach, here we have compared the absolute and relative bark thickness (bark thickness per unit stem diameter) of oak species sharing mixed stands to generalize adaptational significance of bark in relation to fire. For the meta-analysis we considered only natural forests with adequate evidences of fire history. We gave importance to relative bark thickness, which is less sensitive to tree size than absolute thickness while comparing species. At 100 cm girth the average absolute bark thickness was significantly higher for Diploxylon pines (3.65 cm) than for oaks (2.2 cm) and Haploxylon pines (1.65 cm). On an average, the relative bark thickness was higher for the fire adapted Diploxylon pines (6.3%) than for *Quercus* species, (4.4%). The resource allocation efficiency was higher in pine species than for oak species, emphasizing the importance of bark characters in their persistence and range expansion. The difference between species of two principal forest genera in relative bark thickness and hence success in fire affected landscapes across a wide geographical range strongly testifies its adaptational significance to fire.

1. Introduction

Stem bark is considered the main protection against undercanopy or ground surface fires (Lawes et al., 2013; Ferrenberg et al., 2014; Pausas, 2015), though there are several other drivers of variations in bark thickness (Rosell, 2016). An experiment conducted to analyse bark insulating capacity of Alpine tree species by heating bark surface, showed that bark thickness was the main determiner of time required to reach a lethal cambium temperature (Bär and Mayr, 2020). Thick barked species are common in frequently burned areas in all regions where surface fires occur, and geographical variability in bark thickness could be explained on the basis of burned area and precipitation of dry seasons (Pellegrini et al., 2017). As fire frequency and intensity can select for higher investment in bark thickness (Lawes et al., 2013; Pausas, 2015), it can be hypothesized that bark thickness is directly related to fire incidences. As fire frequency increases, relatively thicker bark and a greater rate of bark accumulation should be beneficial for protecting the vascular cambium during fire (Lawes et al., 2011).

Oak (*Quercus*, with > 400 species) and pine (*Pinus*, with 120 species) are the two principal forest forming genera of the extra-tropical latitudes of the Northern Hemisphere, sharing areas in all the three continents. There is a wide overlap in the geographical distribution of oak and pine species throughout the Northern Hemisphere between 20 and 55°N latitudes, with centres in Mexico, USA, Europe, and SE Asia. Oak and pines often form mixed forests in several regions falling in subtropical to temperate and Mediterranean climates. These mixed pine-oak forests (e. g., *Q. cerris*-*P. brutia* and *Q. pubescens* -*P. nigra* in Turkey, Turkmen and Duzenli, 2011; Kaptanoglu and Namli, 2020; *P. strobus* - *Q. alba* in the USA, Abrams et al., 1995; and *P. nigra*-*Q. faginea* in Spain, Gracia et al., 2002) have been affected to a varying extent by fires, land use changes, tree cutting and biomass extraction (Singh et al., 1992; Richardson, 1998; Kelly et al., 2012; Sheffer, 2012).

There occur a wide variety of fire-dependent pine (*Pinus* spp.) forests, both with and without oaks (*Quercus* spp.) and pine savannas across Central America and the Caribbean islands, and extensive open pine forests and woodlands in the tropical and subtropical environments of

* Corresponding author.

E-mail address: surabhigumber19@gmail.com (S. Gumber).<https://doi.org/10.1016/j.foreco.2022.120081>

Received 26 November 2021; Received in revised form 15 January 2022; Accepted 1 February 2022

Available online 10 February 2022

0378-1127/© 2022 Elsevier B.V. All rights reserved.

Southeast and South Asia. In them, frequent fires act as a key ecosystem driver in maintaining stand characteristics (Singh and Singh, 1992; Kraus and Goldammer, 2007; Singh et al., 2016). Fire is an important component of pine and oak forest areas in the Mediterranean regions.

In western and central Himalayan region, frequent human-ignited low intensity fires are common in chir pine (*Pinus roxburghii*) – banj oak (*Quercus leucotrichophora*) forest belt (generally, between 800 and 2000 m altitude) (Singh and Singh, 1986, 1992). There are evidences indicating that fires have changed the regional domination of species and growth forms (Singh and Singh, 1986, 1992; Nautiyal, 2015; Singh et al., 2016). Human-ignited fires have been shaping the Central Himalayan forests from foothills to 1800 m elevation or more for thousands of years, modifying their structural and compositional features (Singh and Singh, 1986). Fire has contributed significantly to the range expansion of several pines both in natural forests and plantations areas (Richardson, 1998).

While the relative success of oak-pine forests depends on several factors and oaks may be favoured in certain conditions (Reyna et al., 2019), bark giving a competitive advantage to species in surface fire affected areas is crucial. However, there can be trade-off between bark thickness and adaptation to other factors, such as drought (De Dios et al., 2018). For example, the carbon allocation to building a thick bark, may limit carbon allocation to drought resistant thick xylem (Fernandes et al., 2008). Partly because of it, conifers generally maintain a wider safety margins, which implies early responses of stomatal closures which reduce hydraulic conductivity before substantial cavitations occur (Tyree and Sperry, 1988). The thick bark may give a competitive advantage to species where fire with low to moderate intensity is common, but such species may be at a disadvantage in a nutrient poor soil because of the limitation of carbon allocation to roots. The thick bark of a canopy species may enable species to dominate a fire affected sites, but it may also eliminate thin-barked under canopy species, and thus reduce species diversity (Strand et al., 2019; Ritcher et al., 2020).

In pine-oak mixed forest stands across the Northern Hemisphere, we recorded the common presence of 45 pine species and 66 oak species (Suppl. mat., Table S1). In the areas where fire is common, Diploxylon pines are the main forest forming species, and often they expand their ranges at the expense of oaks (Richardson, 1998). We hypothesize that since thick bark is a major fire adaptation, the bark of Diploxylon pines should be thicker than of oaks and Haploxylon pines in fire affected areas. While in subalpine forest areas where fire is rare, no such difference between oak and pines should occur, and in such a stressful environment, pine species are expected to be largely of Haploxylon section of *Pinus* genus. The purpose of this investigation, based on a meta-analysis was to examine bark adaptation among the oak and pine species sharing fire affected landscapes across a wide range of environmental conditions in Northern Hemisphere. We considered both absolute bark thickness and relative bark thickness (percentage of stem that is bark, Lawes et al., 2013) while analyzing oaks and pines across their distributional range.

We could procure comparable data on bark thickness of 20 pine species (16 Diploxylon and 04 Haploxylon) and 17 oak species from 92 fire affected sites, while the species numbers are less than we had aspired for, the number of sites and their geographical range somewhat compensated for it.

2. Material and methods

2.1. Study region and species

The study covered much of the global distributional range of mixed pine-oak forests (10 m to 3200 m) (between 120° W and 150° E, and between 10° and 60° N). It included areas of many countries, distributed mainly in regions of temperate, sub-tropical and Mediterranean climates (Fig. 1).

In mixed oak-pine zone of the Northern Hemisphere (Fig. 1) climate

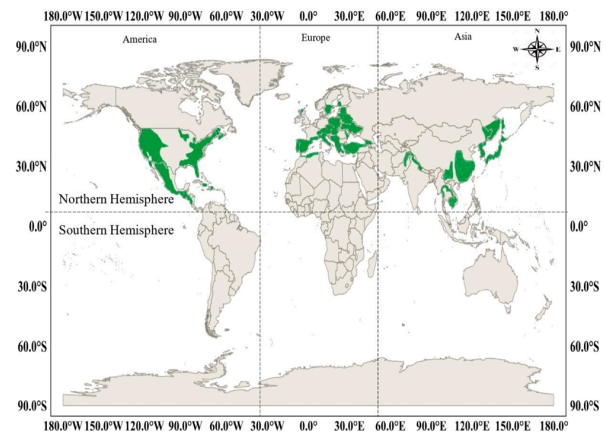


Fig. 1. Global distribution of natural pine-oak mixed forests (based largely on Troup, 1921; Champion and Seth, 1968; Ralhan et al., 1985; Kutiel and Naveh, 1987; Rana et al., 1989; Gallina et al., 1991; Loshali and Singh, 1992; Singh and Singh, 1992; Matlack et al., 1993; Abrams et al., 1995; Cain and Shelton, 1995; Singer et al., 1996; Fule and Covington, 1998; Subedi, 1998; Kaneko and Salamanca, 1999; Kolb and Stone, 2000; Dzwonko and Gawronski, 2002; Beon and Bartsch, 2003; Ford and Brooks, 2003; Kala, 2004; Lee et al., 2004; Rice et al., 2004; Wakeel et al., 2005; Ajbilou et al., 2006; Inclan et al., 2007; Singh, 2010; Slazak et al., 2010; Zavala et al., 2010; Perez-Suarez et al., 2012; Perot et al., 2013; Singh et al., 2016; Bonal et al., 2017; Murata et al., 2017; Barton and Poulos, 2018; Meng et al., 2018; Huo et al., 2019; Zhang et al., 2019; Cruz-Garcia et al., 2020; Kaptanoglu and Namli, 2020; Bowman et al., 2021).

generally ranges from arid to sub-humid with annual precipitation between 200 and 1000 mm. Drought and fires are common (e.g., mean fire return interval being 4.2–17.9 years in SE Arizona, Barton and Poulos, 2018). Lightning-caused surface fires and stand-replacing fires were common in SE Arizona. However, most fire incidences are due to human ignited fires. In Himalayas most fires are human-ignited and occur during pre-monsoon period (March-May) which is warm and dry (Singh et al., 2016).

The species for which bark data were available accounted for 44.4% of pine species (20 of 45, 16 Diploxylon and 4 Haploxylon pines) and 25.8% of oak species (17 of 66) occurring in pine-oak mixed forests. Most study species were > 20 m tall canopy species. Of the six Himalayan pines four species, *P. kesiya*, *P. roxburghii*, *P. gerardiana*, and *P. wallichiana* form association with *Quercus* species (*Q. ilex*, *Q. fenes-trata*, *Q. leucotrichophora*, *Q. serrata*) (Table 1). Of them, *P. roxburghii*, and *P. kesiya*, the Diploxylon pines are known to have evolved in relation to fire, whereas the others, which were Haploxylon pines (*P. gerardiana*, and *P. wallichiana*) are fire sensitive and adapted to abiotically stressed environment, such as dry and cold areas, including timberlines (Singh et al., 2018; Singh et al., 2021). The most common and widely described pine-oak association in Himalayas is of chir (*P. roxburghii*) and banj-oak (*Q. leucotrichophora*) association, but several other combinations also occur (Table 1).

2.2. Methods

We did an extensive literature survey to collect data and information to analyze spatial distribution of pine-oak mixed forests of the world, their morphological characters, adaptation to fire, and disturbance, resulting from biomass collection and grazing, abiotic stress: Troup, 1921; Champion and Seth, 1968; Ralhan et al., 1985; Kutiel and Naveh, 1987; Haigh et al., 1989; Rana et al., 1989; Gallina et al., 1991; Biondi et al., 1992; Loshali and Singh, 1992; Singh and Singh, 1992; Matlack et al., 1993; Abrams et al., 1995; Cain and Shelton, 1995; Cain, 1996; Singer et al., 1996; Compton et al., 1998; Fule and Covington, 1998; Subedi, 1998; Kaneko and Salamanca, 1999; Kolb and Stone, 2000;

Table 1

Globally reported pine-oak associations used in the present study. The bark thickness values given for the species in parentheses for each species are based on Troup, 1921; Champion and Seth, 1968; Ralhan et al., 1985; Rana et al., 1989; Biondi et al., 1992; Loshali and Singh, 1992; Matlack et al., 1993; Abrams et al., 1995; Cain and Shelton, 1995; Cain, 1996; Compton et al., 1998; Fule and Covington, 1998; Kolb and Stone, 2000; Park, 2001; Sultan et al., 2001; Gracia et al., 2002; Kala, 2004; Zavala and Zea, 2004; Wakeel et al., 2005; Kumar et al., 2016; Inclan et al., 2007; Perez-Suarez et al., 2009; Sen and Gok, 2009; Clark et al., 2010; Copenheaver et al., 2010; Gurarni et al., 2010; Majila and Kala, 2010; Sharma and Singh, 2010; Slazak et al., 2010; Joshi and Bhatt, 2011; Turkmen and Duzenli, 2011; Chaturvedi et al., 2012; Perez-Suarez et al., 2012; Bieng et al., 2013; Perot et al., 2013; George et al., 2014; Perez-Suarez et al., 2014; Khan et al., 2015; Castillo et al., 2016; Gill-tena et al., 2016; Shahabuddin et al., 2017; Singh et al., 2016; Bonal et al., 2017; Lal et al., 2017; Bargali et al., 2018; Barton and Poulos, 2018; Meng et al., 2018; Kumar and Kumari, 2019; Woziwoda et al., 2019).

Pines	Oaks	Country
<i>Pinus echinata</i> Mill. (2.08 cm)	<i>Quercus alba</i> L. (0.84 cm); <i>Q. marilandica</i> Muenchh. (0.48 cm)	USA
<i>P. ponderosa</i> Douglas ex C. Lawson (7.62 cm)	<i>Q. gambelii</i> Nutt. (1.55 cm)	USA
<i>P. rigida</i> Mill. (3.04 cm)	<i>Q. alba</i> L. (0.84 cm); <i>Q. marilandica</i> Muenchh. (0.48 cm)	USA
<i>P. strobus</i> L. (3.18 cm)	<i>Q. alba</i> L. (0.84 cm); <i>Q. rubra</i> L. (0.84 cm)	USA
<i>P. taeda</i> L. (3.51 cm)	<i>Q. alba</i> L. (0.84 cm); <i>Q. marilandica</i> Muenchh. (0.48 cm)	USA
<i>P. virginiana</i> Mill. (1.27 cm)	<i>Q. alba</i> L. (0.84 cm); <i>Q. rubra</i> L. (0.84 cm)	USA
<i>P. arizonica</i> Engelm. (5 cm)	<i>Q. arizonica</i> Sarg. (2.5 cm); <i>Q. laeta</i> Liebm. (0.25 cm)	USA; Mexico
<i>P. engelmannii</i> Carr. (4.44 cm)	<i>Q. arizonica</i> Sarg. (2.5 cm); <i>Q. laeta</i> Liebm. (0.25 cm)	USA; Mexico
<i>P. leiophylla</i> Schiede ex Schltdl. & Cham. (6.98 cm)	<i>Q. arizonica</i> Sarg. (2.5 cm); <i>Q. laeta</i> Liebm. (0.25 cm)	USA; Mexico
<i>P. cembroides</i> Zucc. (1.27 cm)	<i>Q. potosina</i> Trel. (0.23 cm)	Mexico
<i>P. halepensis</i> Mill. (3.81 cm)	<i>Q. ilex</i> L. (0.64 cm); <i>Q. suber</i> L. (8.9 cm)	Portugal; Spain (Mediterranean)
<i>P. pinea</i> L. (3.3 cm)	<i>Q. ilex</i> L. (0.64 cm); <i>Q. suber</i> L. (8.9 cm)	Spain (Mediterranean)
<i>P. nigra</i> J.F. Arnold (3.1 cm)	<i>Q. coccifera</i> L. (0.7 cm); <i>Q. faginea</i> Lam. (2.2 cm); <i>Q. ilex</i> L. (0.64 cm); <i>Q. suber</i> L. (8.9 cm)	Spain (Mediterranean); Turkey
<i>P. sylvestris</i> L. (3.18 cm)	<i>Q. petraea</i> (Matt.) Liebl. (1.9 cm); <i>Q. rubra</i> L. (0.84 cm); <i>Q. ilex</i> L. (0.64 cm); <i>Q. suber</i> L. (8.9 cm)	France; Spain (Mediterranean); Germany; Poland
<i>P. pinaster</i> Aiton (3.18 cm)	<i>Q. ilex</i> L. (0.64 cm)	Italy
<i>P. brutia</i> Ten. (4.44 cm)	<i>Q. cerris</i> L. (5 cm); <i>Q. coccifera</i> L. (0.7 cm)	Turkey
<i>P. gerardiana</i> Wall. ex D. Don (1.56 cm)	<i>Q. ilex</i> L. (0.64 cm)	Pakistan; India
<i>P. roxburghii</i> Sarg. (3.5 cm)	<i>Q. leucotrichophora</i> A. Camus (1.42 cm); <i>Q. glauca</i> Thunb. (0.41 cm)	Pakistan; India
<i>P. wallichiana</i> A. B. Jacks. (2.05 cm)	<i>Q. ilex</i> L. (0.64 cm); <i>Q. leucotrichophora</i> A. Camus (1.42 cm)	Pakistan; India
<i>P. kesiya</i> Royle ex Gordon (3.5 cm)	<i>Q. serrata</i> Murray. (1.91 cm); <i>Q. fenestrata</i> syn. <i>Lithocarpus</i> <i>fenestrata</i> (Roxb.) Rehder (1.27 cm)	India

Park, 2001; Silva and Rodriguez, 2001; Sultan et al., 2001; Beon and Bartsch, 2003; Ford and Brooks, 2003; Naing, 2003; Kala, 2004; Lee et al., 2004; Nagendra et al., 2004; Rice et al., 2004; Wakeel et al., 2005; Ajbilou et al., 2006; Inclan et al., 2007; Zavala et al., 2010; Kumar et al., 2016; Meng et al., 2018; Shalizi et al., 2018; Huo et al., 2019; Kumar and Kumari, 2019; Reyna et al., 2019; Woziwoda et al., 2019; Zhang et al., 2019; Cruz-Garcia et al., 2020; Gea-Izquierdo et al., 2020; Kaptanoglu and Namli, 2020; Bouwman et al., 2021.

For selecting data/information for our meta-analysis, we followed certain criteria: (i) Only fire affected sites having both oak and pine species were selected. In North America, particularly Mexico generally a pine species is associated with three to four oak species, while in Himalayas generally the sites had one species of each; (ii) We considered only natural forest landscape with recorded fire histories, and avoided plantations; (iii) Stands had fair representation of mature trees with stem girth > 100 cm.

The data on bark thickness, tree height, diameter (at breast height, 1.37 m) were collected from Troup, 1921; Gamble, 1985; Singh and Singh, 1992; Richardson, 1998; Catry et al., 2010; Sen et al., 2011; Saladin, 2013; Thomas and Bennett, 2014; Gričar et al., 2015; Hammond et al., 2015; Pimont et al., 2018; Gumber et al., 2021. Bark thickness data were not available for all of the globally recorded tree species of pine-oak associations. Since we have been associated with Himalayan region for quite some time, examples from the region are frequently referred to in the paper.

We used polynomial regression to analyze the relationship between stem diameter and bark thickness of a species. Linear regression was used to compare the bark thickness between pines and oaks occurring together in mixed forests plotted in relation to longitude in the Northern Hemisphere. Relative bark thickness was calculated for each individual as the ratio of bark thickness to diameter. By assuming that the cost of thick bark is a function of the cross-sectional area of bark, the relative costs of bark growth among individuals with different relative bark thickness was calculated by:

$$BA = \pi r^2 - \pi(r - BT)^2$$

where BA, cross-sectional area of the bark (cm²); r, radius of the stem (cm); BT, bark thickness (cm).

Polynomial regression was conducted to test the relationship between cross sectional bark area and stem diameter, and stem diameter and relative bark thickness. Differences among species in bark thickness, relative bark thickness, and the time required to kill the cambium were analysed with one-way ANOVA with species as a fixed factor. The significance of difference in relative bark thickness between Diploxylon and Haploxylon pines, Diploxylon and *Quercus* species, Haploxylon pines and *Quercus* species was tested at 5% level of significance using the independent samples *t*-test with the help of SPSS 20.0 software.

3. Results

3.1. Composition of mixed pine and oak forests

Out of 114 sites where oak and pines formed mixed forests in Northern Hemisphere, Diploxylon pine and oak species shared forests at 75 sites, and Haploxylon pine and oak species at 20 sites, and species of the three groups at 19 sites. Percentage of sites for Diploxylon and Haploxylon pines match their species numbers (about 2/3rd of the sites were under Diploxylon pines which matches with their percentage of species numbers). In Himalayas, sites having species of all the three groups are not found as Diploxylon and Haploxylon pines are attitudinally separated. Diploxylon pines occur in lower elevations (<2000 m) with moderate favorable temperature and monsoon precipitation while Haploxylon pines occupy higher ranges (between 2000 and 3500 m) in relatively dry and cold areas including inner valleys where monsoon precipitation hardly penetrates.

3.2. Absolute bark thickness

From the west to east the stem bark thickness tended to decline in pines and increased in oaks, thus resulting in narrowing down of the gap between them (Fig. 2). Between the pairs of species of Diploxyton pines and oaks, the bark thickness was generally considerably thicker for the former, and between the two pine groups, it was markedly thinner in Haploxyton pines (Fig. 2).

Though only four Haploxyton pines occurred in fire affected landscapes, their average bark thickness (2.02 cm) was very similar to that of the Haploxyton pines as whole (2.10 cm), indicating that they faithfully represented the entire Haploxyton group. The average bark thickness of Diploxyton pines of our meta-analysis was (3.87 cm), marginally higher than the average of the group as a whole (3.42 cm), possibly because our sites were more frequently burned than the overall Diploxyton pines.

3.3. Bark thickness in relation to tree diameter

The stem bark thickness varied greatly within and between *Pinus* and *Quercus* species. In Diploxyton pines, Haploxyton pines and oaks the stem bark thickness ranged from 1.27 to 7.62 cm, 1.27 to 3.18 cm, and 0.23 to 8.9 cm, respectively. There was a weak tendency of increase in bark thickness with increase in stem diameter within each of the three groups: Diploxyton pines, $R^2 = 0.35$; $p > 0.01$; Haploxyton pines, $R^2 = 0.88$; $p > 0.01$, and *Quercus* species, $R^2 = 0.07$; $p > 0.01$ (Fig. 3). At about 100 cm stem girth, the bark thickness was 3.65 cm in Diploxyton pines, 1.56 cm in Haploxyton pines and 2.20 in *Quercus* species. Among oak species, bark was generally thin, below 2 cm, except in *Q. suber* (8.9 cm) and *Q. cerris* (5 cm) which are fire tolerant species. *Q. suber*, the cork oak forests cover about 2.2 million ha, mostly in the Mediterranean region. Occurring in open woodlands of dry climate where fire is an ever-present hazard, Diploxyton pines had about 48% thicker bark than Haploxyton pines and 52.8% thicker than *Quercus* species. With regard to height in relation to stem diameter, the oaks and Diploxyton pines were similar, and had higher values than the Haploxyton pines in all the three groups, bark thickness was significantly different ($F_{2,34} = 5.42$, $p < 0.01$).

Relative bark thickness (bark thickness per unit stem diameter) was significantly different among Diploxyton and Haploxyton pines ($t = 2.76$, $p < 0.05$), however, the difference was non-significant between Diploxyton pines and *Quercus* species and Haploxyton pines and *Quercus* species. On an average, relative bark thickness was higher for Diploxyton pines (6.3%) than for *Quercus* species, 4.4% and Haploxyton pines, 3.6% (Fig. 4). A comparison between a Himalayan oak (*Q. leucotrichophora*) and pine (*P. roxburghii*) indicated that while in both relative bark thickness decreased from young tree stage to mature tree

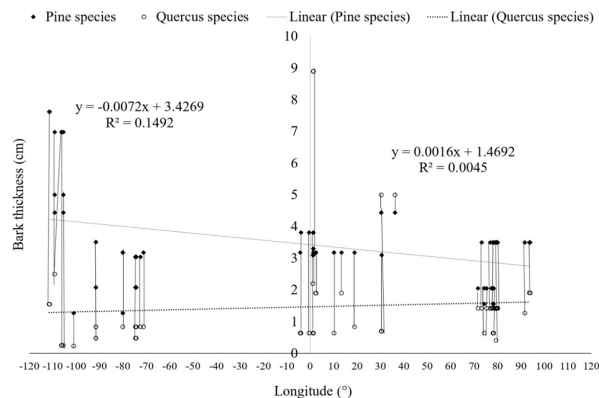


Fig. 2. Comparison of bark thickness between pine and oak species occurring together (indicated in pairs) in mixed forests plotted in relation to longitude in the Northern Hemisphere.

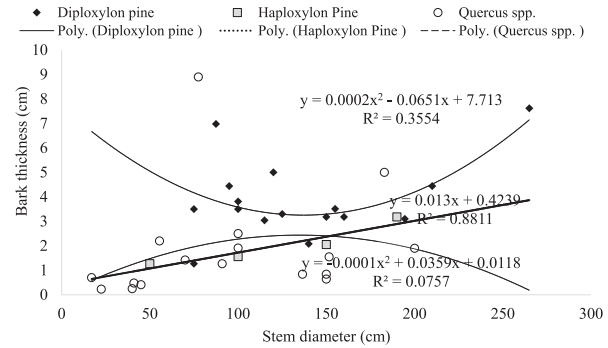


Fig. 3. Relationship between stem diameter (cm) and bark thickness (cm) for all the studied pine-oak associations.

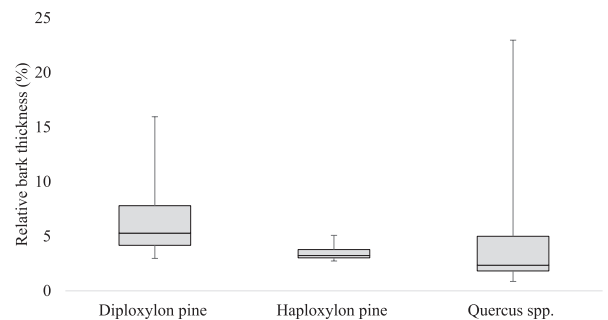


Fig. 4. Boxplots of relative bark thickness (%) for studied three groups.

stage, the decline was sharper in the pine (from 18% to 7%) than the oak (from 12% to 8%). The pine allocated more carbon to bark when the individuals were young and particularly susceptible to top kill by fire. At later stage when trees were too tall to be affected by fire, it reduced bark allocation. The cross sectional area of bark, which represents the cost, increases rapidly with diameter (Lawes et al., 2013).

Relative bark thickness declined with stem diameter for all species (Fig. 5), however, the rate of decline with stem diameter was higher in Diploxyton pines than Haploxyton pines and *Quercus* species.

In all the three groups cross sectional bark area increased significantly with diameter ($R^2 = 0.70$; $p < 0.05$; $R^2 = 0.91$; $p < 0.05$, and $R^2 = 0.33$; $p < 0.05$ respectively for Diploxyton pines, Haploxyton pines and oaks) (Fig. 6). The difference between the two species for bark cross sectional area in relation to diameter increased with stem diameter.

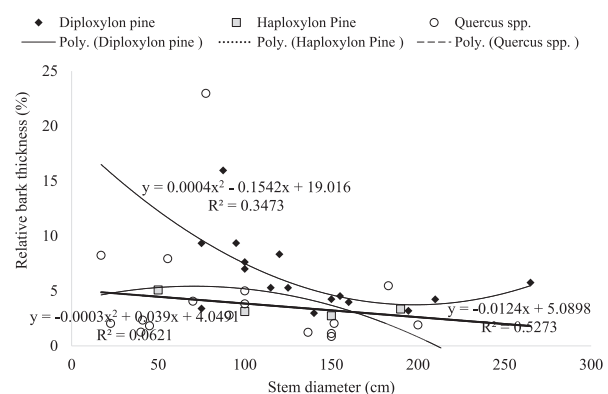


Fig. 5. Relationship between stem diameter (cm) and relative bark thickness (%) of Diploxyton pines, Haploxyton pines and *Quercus* species.

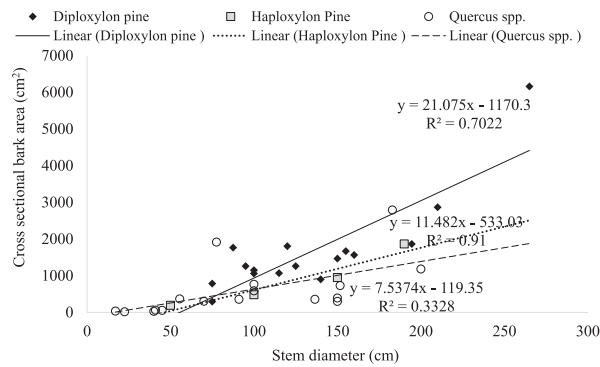


Fig. 6. Relationship between stem diameter (cm) and cross sectional bark area (cm^2) of Diploxylon pines, Haploxylon pines and *Quercus* species.

4. Discussion

The present study compares bark thickness of oak and pine species occurring together in fire affected locations over a wide geographical range from Mexico and the USA on one end to the eastern Himalaya on the other. We addressed two questions: first, what is the position of two important forest forming genera, oak and pine with regard to bark thickness in fire (generally, small and frequent surface fires) affected landscapes over a wide range of environmental conditions in the Northern Hemisphere; second, to what extent investment in bark development in species groups changes with tree diameter across their species. In other words, how efficiently they make use of allocation of resources to bark in relation to trade-offs.

That the pine species of Diploxylon and Haploxylon lineages did not occur together in 83.3% of 114 sites for which data were compiled, emphasizes their spatial separation is consistent with in that their specific adaptational traits (Keeley et al., 2012). Since Diploxylon pines are adapted to forest fires which are common in relatively more productive areas and Haploxylon pines to abiotic stress, they are spatially separated (Singh et al., 2021). As predicted, we found that Diploxylon pines, which are known to be fire-resistant (Keeley et al., 2012; Singh et al., 2018), had significantly thicker bark, hence proportionately more investment in bark than Haploxylon pine and oak species. So the difference between pine and oak species was in (i) absolute total bark thickness, (ii) in proportional resource allocation to bark (relative bark thickness) and allocation of resources to bark at an early stage, when individuals are most vulnerable to fire. In each Diploxylon pines are more equipped to deal with surface fire regime than oak species. However, unlike Haploxylon pines, which are highly fire sensitive (Keeley et al., 2012), oak species vary to an extent in fire protection, and some of them invest substantially in stem protection near the ground where fire damage is most likely (Pausas, 2015) and several oaks are effective resprouters (Schwilk et al., 2016). As an example, a study on Floridian oaks showed that the outer bark thickness was more for fire-resistant oak species (0.60–1.07 cm) than for fire-sensitive oak species (0.33–0.84 cm) (Adams and Jackson, 1995). *Q. lanata*, which occurs frequently in fire affected sites has relatively thicker bark (>2 cm) (Tewari et al., 1989). Among the oaks, cork oak, *Q. suber* and *Q. cerris* have exceptionally thick bark (8.9 and 5 cm respectively).

In Himalayas, oaks in general have significantly thinner bark (mean, 1.13 cm) and relative bark thickness (0.27 mm cm^{-1}) than Diploxylon pines (mean, 3.5 cm thick bark) growing with them. Himalayan oaks growing with pines, particularly *P. roxburghii* are adversely affected by frequent surface fires, though they are able to survive a few fire events, by resprouting (Troup, 1921; Ralhan et al., 1982; Nautiyal, 2015). However, as light availability increases because of the disturbance that fires cause, the strong light demanding *P. roxburghii* outcompetes oaks like *Q. leucotrichophora* and *Q. floribunda* (Singh and Singh, 1986). Thus,

the range expansion of *P. roxburghii* in western Himalayas at the expense of oaks is because of effective bark protection against mild surface fires as well as increased light availability that the fire creates.

The thick bark development has a considerable cost in terms of resource allocation, as well as in opportunity costs caused by limiting diffusion of water, oxygen and light through stems (Cavaleri et al., 2006; Teskey et al., 2008). A thick bark could result in a slower height growth thus jeopardizing trees escape from fire flames. Difference in bark thickness counts more where a small difference in bark thickness makes a large difference in fitness. Such a fire regime is characterized by low intensity, and frequent surface fires (Pausas, 2015). In areas, where fires are very intensive, trees regardless of bark thickness may be destroyed, so thick bark stems are not selected (Pausas, 2015).

The cost of thick bark in a fire affected landscape could be extended to community level. In Himalayas, forest fires are frequent patchy, small and mild and about 80% of fire incidences occurs during pre-monsoon period (March-May), when conditions were dry, and leaf litter fall takes place to fires (Singh et al., 2016). Such a fire regime has given competitive advantage to Diploxylon pines over oaks as the pines develop a thick bark at early age. However, frequent small fires lead to the local extinction of thin barked undercanopy species like *Myrica esculenta* (bark 0.48 cm thick) and *Litsea umbrosa* (bark 0.55 cm thick). The relatively thick barked (1.23 cm) *Rhododendron arboreum* survives but not for long (Troup, 1921; Singh and Singh, 1992; Gumber et al., 2021). The shrub density in *P. roxburghii* forest of Uttarakhand Himalaya is reduced to 2.7 individuals/100 m^2 compared to 257.4 individuals/100 m^2 in adjacent *Q. leucotrichophora* forests (Singh and Singh, 1992). However, fire has promoted many grasses, and when disturbances are moderate it may lead to increased species richness (Richter et al., 2019; Strand et al., 2019).

Oaks do not grow rapidly in height, hence cannot employ the height escape strategy of fire as do some eucalypts in savannas. Being a good sprouter, oaks growing in a surface fire regime can produce a good crop of saplings. Oaks of surface fire regimes have a fairly thick bark, but not as thick as pines (Pausas, 2015). However, as a genus *Quercus* varies widely with regard to the association of its species with fire, which can occur in savannas with frequent fire regime to forests with well-developed canopy cover and limited fire incidences (Adams and Jackson, 1995). In Mexican pine-oak forests frequent fires are essential for the domination of pine over oaks (Fule and Covington, 1998). Species can have other fire adaptation than bark thickness, like sprouting and serotiny. *Q. suber* has a corky bark as well as its ability to sprout, while *P. canariensis* has serotiny in addition to thick corky bark (Climent et al., 2004). Because plant employ several adaptational fire resistance means, the relationship between bark thickness and fire resistance may not always strong. Then, several other factors like drought can drive bark thickness as bark is multifunctional in nature (Rosell, 2016). Nevertheless, the linkage between consistently higher investment in bark in Diploxylon pines and their expansion among fire affected forests at the cost of oaks across a wide range of forest communities in the Northern Hemisphere signify the importance of stem bark as an adaptation to fire. The concept of the pines persistence despite the cretaceous diversification of angiosperms (Keeley et al. 2012; Singh et al. 2018) can be analyzed more effectively with focus on oaks, with which pines shared dominance.

5. Conclusions

Our meta-analysis shows that pines have greater absolute and relative bark thickness (i.e. relative to stem diameter or bark area as percentage of stem, Hofmann et al., 2003) values than oaks. Absolute bark thickness is a good indicator of the protective function of bark in relation to fire, but heavily depends on stem size, while relative bark thickness take care of stem size variation (see also Lawes et al., 2013). Generally, allocation to bark begins at juvenile stage of tree species emphasizing a kind of “commitment” of the species to fire adaption (Lawes et al.,

2013). However, there are trade-offs for example more carbon allocation to stem bark may leave pines with inadequate carbon to develop the cavitation-resistant xylem tissues. Lignin accounts for varying but considerable percentage of bark (13–43%) and wood (15–36%) of trees (Fengel and Wegener, 1983). In a majority of oaks, wood density is between 600 and 900 kg m⁻³, compared to 400–650 kg m⁻³ for pines. The relatively thin bark of oaks may partly reflect the trade-off effect. Adaptation to fire and drought thus has separate lignin cost. To deal with this, pines and other conifers have developed wider hydraulic safety margins, which imply an early response of stomatal closure, resulting in a reduced hydraulic conductivity, well before it reaches the damaging point (Tyree and Sperry, 1988; Choat et al., 2012). While this water conservation mechanism protects pines from hydraulic failures, it deprives them from carbon gains in favorable conditions. In contrast, several oaks keep stomata open even at 30–40% lower predawn water potential than pines (Singh et al., 2006), thus procuring carbon even during unfavorable seasons. There is a trade-off also at a community scale, the fire tolerant bark of pines enables them to form monodominant stands resulting in local extinction of thin barked tree species, and most shrubs.

CRedit authorship contribution statement

Ripu Daman Singh: Data curation, Writing – original draft, Methodology, Software. **Surabhi Gumber:** Visualization, Investigation, Writing – original draft. **Harshita Joshi:** Visualization, Investigation. **Surendra P. Singh:** Conceptualization, Writing – original draft, Supervision.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

All the authors acknowledge the research facilities provided by Central Himalayan Environment Association (CHEA), Nainital (Uttarakhand), India and Indian National Science Academy to provide Honorary Scientist Fellowship to one of the author (SPS). Surabhi Gumber, and Ripu Daman Singh are grateful to Department of Forestry and Environmental Science, DSB Campus, Kumaun University for liberal support. Harshita Joshi is thankful to Department of Botany, IPGGPG College of Commerce, Haldwani, Kumaun University, Nainital for liberal support.

Appendix A. Supplementary material

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.foreco.2022.120081>.

References

Abrams, M.D., Orwig, D.A., Demeo, T.E., 1995. Dendroecological analysis of successional dynamics for a presettlement-origin white-pine-mixed-oak forest in the southern Appalachians, USA. *J. Ecol.* 83 (1), 123. <https://doi.org/10.2307/2261156>.
 Adams, D.C., Jackson, J.F., 1995. Estimating the allometry of tree bark. *Am. Midl. Nat.* 134 (1), 99. <https://doi.org/10.2307/2426487>.
 Ajilou, R., Maranon, T., Arroyo, J., 2006. Ecological and biogeographical analyses of Mediterranean forests of northern Morocco. *Acta Oecol.* 29, 104–113. <https://doi.org/10.1016/j.actao.2005.08.006>.
 Bär, A., Mayr, S., 2020. Bark insulation: Ten Central Alpine tree species compared. *For. Ecol. Manage.* 474, 118361. <https://doi.org/10.1016/j.foreco.2020.118361>.
 Bargali, K., Manral, V., Padalia, K., Bargali, S.S., Upadhyay, V.P., 2018. Effect of vegetation type and season on microbial biomass carbon in Central Himalayan forest soils, India. *Catena* 171, 125–135.

Barton, A.M., Poulos, H.M., 2018. Pine vs. oaks revisited: Conversion of Madrean pine-oak forest to oak shrubland after high-severity wildfire in the Sky Islands of Arizona. *For. Ecol. Manage.* 414, 28–40. <https://doi.org/10.1016/j.foreco.2018.02.011>.
 Beon, M.S., Bartsch, N., 2003. Early seedling growth of pine (*Pinus densiflora*) and oaks (*Quercus serrata*, *Q. mongolica*, *Q. variabilis*) in response to light intensity and soil moisture. *Plant Ecol.* 167, 97–105.
 Bieng, M.A.N., Perot, T., Coligny, F.D., Goreaud, F., 2013. Spatial pattern of trees influences species productivity in a mature oak-pine mixed forest. *Eur. J. Forest Res.* 132, 841–850.
 Biondi, F., Klemmedson, J.O., Kuehl, R.O., 1992. Dendrochronological analysis of single-tree interactions in mixed pine-oak stands of central Arizona, USA. *For. Ecol. Manage.* 48 (3–4), 321–333.
 Bonal, D., Pau, M., Toigo, M., Granier, A., Perot, T., 2017. Mixing oak and pine trees does not improve the functional response to severe drought in central French forests. *Ann. For. Sci.* 74, 72. <https://doi.org/10.1007/s13595-017-0671-9>.
 Bouwman, M., Forrester, D.I., den Ouden, J., Nabuurs, G.-J., Mohren, G.M.J., 2021. Species interactions under climate change in mixed stands of Scots pine and pedunculate oak. *For. Ecol. Manage.* 481, 118615. <https://doi.org/10.1016/j.foreco.2020.118615>.
 Cain, M.D., 1996. Hardwood snag fragmentation in a Pine-oak forest of Southeastern Arkansas. *Am. Midl. Nat.* 136 (1), 72. <https://doi.org/10.2307/2426632>.
 Cain, M.D., Shelton, M.G., 1995. Thirty-eight years of autogenic, woody understory dynamics in a mature, temperate pine-oak forest. *Can. J. For. Res.* 25 (12), 1997–2009.
 del Castillo, J., Comas, C., Voltas, J., Ferrio, J.P., 2016. Dynamics of competition over water in a mixed oak-pine Mediterranean forest: spatio-temporal and physiological components. *For. Ecol. Manage.* 382, 214–224.
 Catry, F.X., Rego, F., Moreira, F., Fernandes, P.M., Pausas, J.G., 2010. Post-fire tree mortality in mixed forests of central Portugal. *For. Ecol. Manage.* 260 (7), 1184–1192.
 Cavaleri, M.A., Oberbauer, S.F., Ryan, M.G., 2006. Wood CO₂ efflux in a primary tropical rain forest. *Glob. Change Biol.* 12, 2442–2458.
 Champion, H.G., Seth, S.K., 1968. A Revised Survey of the Forest Types of India. Government of India Press, New Delhi, p. 403.
 Chaturvedi, S., Tewari, V., Sharma, S., Oehl, F., Wiemken, A., Prakash, A., Sharma, A.K., 2012. Diversity of Arbuscular Mycorrhizal Fungi in Oak-pine forests and agricultural land prevalent in the Kumaun Himalaya Hills, Uttarakhand, India. *Br. Microbiol. Res. J.* 2, 82–96.
 Choat, B., Jansen, S., Brodribb, T.J., Cochard, H., Delzon, S., Baskar, R., Bucci, S.J., Feild, T.S., Gleason, S.M., Hacke, U.G., Jacobsen, A.L., Lens, F., Maherali, H., Martínez-Vilalta, J., Mayr, S., Mencuccini, M., Mitchell, P.J., Nardini, A., Pittermann, J., Pratt, R.B., Sperry, J.S., Westoby, M., Wright, I.J., Zanne, A.E., 2012. Global convergence in the vulnerability of forests to drought. *Nature* 491 (7426), 752–755. <https://doi.org/10.1038/nature11688>.
 Clark, K.L., Skowronski, N., Hom, J., 2010. Invasive insects impact forest carbon dynamics. *Glob. Change Biol.* 16 (1), 88–101.
 Climent, J., Tapias, R., Pardos, J.A., Gil, L., 2004. Fire adaptations in the Canary Islands pine (*Pinus canariensis*). *Plant Ecol.* 171 (1/2), 185–196.
 Compton, J.E., Boone, R.D., Motzkin, G., Foster, D.R., 1998. Soil carbon and nitrogen in a pine-oak sand plain in central Massachusetts: role of vegetation and land-use history. *Oecologia* 116 (4), 536–542.
 Copenheaver, C.A., Gartner, H., Schafer, I., Vaccari, F.P., Cherubini, P., 2010. Drought-triggered false ring formation in a Mediterranean shrub. *Botany* 88, 545–555.
 Cruz-Garcia, F., Monárrez González, J.C., Tecle, A., Wehenkel, C., Perez-Verdin, G., Van Stan, J.T., 2020. Effects of stand variables on stemflow and surface runoff in pine-oak forests in northern Mexico. *PLoS One* 15 (6), e0235320.
 De Dios, V.R., Arteaga, C., Hedo, J., Gil-Pelegrín, E., Voltas, J., 2018. A trade-off between embolism resistance and bark thickness in conifers: are drought and fire adaptations antagonistic? *Plant Ecol. Div.* 11 (3), 253–258. <https://doi.org/10.1080/17550874.2018.1504238>.
 Dzwonko, Z., Gawronski, S., 2002. Effect of litter removal on species richness and acidification of a mixed oak-pine woodland. *Biol. Conserv.* 106 (3), 389–398.
 Fengel, D., Wegener, G., 1983. Wood: chemistry, ultrastructure, reactions. Berlin, New York. <https://doi.org/10.1515/9783110839654.fm>.
 Fernandes, P.M., Vega, J.A., Jiménez, E., Rigolot, E., 2008. Fire resistance of European pines. *For. Ecol. Manage.* 256 (3), 246–255. <https://doi.org/10.1016/j.foreco.2008.04.032>.
 Ferrenberg, S., Mitton, J.B., Jones, H., 2014. Smooth bark surfaces can defend trees against insect attack: resurrecting a 'slippery' hypothesis. *Funct. Ecol.* 28 (4), 837–845.
 Ford, C.R., Brooks, J.R., 2003. Hydrological and climatic responses of *Pinus elliottii* var. *densa* in mesic pine flatwoods Florida, USA. *Ann. For. Sci.* 60 (5), 385–392. <https://doi.org/10.1051/forest:2003030>.
 Fule, P.Z., Covington, W.W., 1998. Spatial patterns of Mexican pine-oak forests under different recent fire regimes. *Plant Ecol.* 134, 197–209.
 Gallina, S., Galina-Tessaro, P., Alvarez-Crdenas, S., 1991. Mule deer density and pattern distribution in the pine-oak forest at the Sierra de La Laguna in Baja California Sur, Mexico. *Ethol. Ecol. Evol.* 3 (1), 27–33. <https://doi.org/10.1080/08927014.1991.9525386>.
 Gamble, J.S., 1985. A manual of Indian timbers: an account of the growth, distribution, and uses of the trees and shrubs of India and Ceylon, with descriptions of their wood-structure. Hind edition. International Book Distributors, Dehra Dun, India.
 Gea-Izquierdo, G., Aranda, I., Canellas, I., Dorado-Linan, I., Olano, J.M., Benito, D.M., 2020. Contrasting species decline but high sensitivity to increasing water stress on mixed pine-oak ecotone. *J. Ecol.* 109, 109–124.

- George, R., Padalia, H., Kushwaha, S.P.S., 2014. Forest tree species discrimination in western Himalaya using EO-1 Hyperion. *Int. J. Appl. Earth Obs. Geoinf.* 28, 140–149.
- Gil-Tena, A., Aquilué, N., Duane, A., De Cáceres, M., Brotons, L., 2016. Mediterranean fire regime effects on pine-oak forest landscape mosaics under global change in NE Spain. *Eur. J. Forest Res.* 135 (2), 403–416.
- Gracia, M., Retana, J., Roig, P., 2002. Mid-term successional patterns after fire of mixed pine-oak forests in NE Spain. *Acta Oecol.* 23 (6), 405–411.
- Grčar, J., Jagodic, Š., Prislán, P., 2015. Structure and subsequent seasonal changes in the bark of sessile oak (*Quercus petraea*). *Trees* 29 (3), 747–757. <https://doi.org/10.1007/s00468-015-1153-z>.
- Gumber, S., Singh, R.D., Ram, J., Tewari, A., Singh, S.P., 2021. Bark thickness analysis of four dominant forest trees species of Central Himalayan forests varying in exposure to surface fires. *Trees*. <https://doi.org/10.1007/s00468-021-02239-z>.
- Gurarni, D., Arya, N., Yadava, A., Ram, J., 2010. Studies on plant biodiversity of pure *Pinus roxburghii* Sarg. Forest and mixed pine-oak forest in Uttarakhand Himalaya. *N. Y. Sci. J.* 3, 1–5.
- Haigh, M.J., Rawat, J.S., Bartarya, S.K., 1989. Environmental indicators of Landslide activity along the Kilbury road, Nainital, Kumaun Lesser Himalaya. *Mt. Res. Dev.* 9 (1), 25. <https://doi.org/10.2307/3673462>.
- Hammond, D.H., Varner, J.M., Kush, J.S., Fan, Z., 2015. Contrasting sapling bark allocation of five southeastern USA hardwood tree species in a fire prone ecosystem. *Ecosphere* 6 (7), art112. <https://doi.org/10.1890/ES15-00065.1>.
- Huo, X., Wang, D., Bing, D., Li, Y., Kang, H., Yang, H., Wei, G., Chao, Z., 2019. Appropriate removal of forest litter is beneficial to *Pinus tabulaeformis* Carr. regeneration in a pine and oak mixed forest in the Qinling Mountains, China. *Forests* 10, 1–12. <https://doi.org/10.3390/f10090735>.
- Inclán, R., Torre, D.D., Benito, M., Rubio, A., 2007. Soil CO₂ efflux in a mixed pine-oak forest in Valsain (Central Spain). *Sci. World J.* 7, 166–174.
- Joshi, K., Bhatt, D., 2011. Birds of three different forest habitats in Nainital district (Western Himalaya), Uttarakhand, India. *Indian Birds* 7, 33–37.
- Kala, C.P., 2004. Indigenous uses and structure of chir pine forest in Uttaranchal Himalaya, India. *Int. J. Sustain. Dev. World Ecol.* 11 (2), 205–210.
- Kaneko, N., Salamanca, E., 1999. Mixed leaf litter effects on decomposition rates and soil microarthropod communities in an oak-pine stand in Japan. *Ecol. Res.* 14 (2), 131–138.
- Kaptanoglu, A.S., Namli, A., 2020. Wildfire and harvesting effects on carbon dynamics in an oak-pine mixed forest. *Biogeosci. Forestry* 13, 435–440. <https://doi.org/10.3832/for3312-013>.
- Kelly, M.J., Betsch, J., Wulsch, C., Mesa, B., Mills, L.S., 2012. Noninvasive sampling for carnivores. In: Boitani, L., Powell, R. (Eds.), *Carnivore ecology and conservation: A hand book of techniques*. Oxford University Press, pp. 47–69.
- Khan, H., Akbar, M., Zaman, M., Ali, A., Hyder, S., Khan, M., Nafees, M.A., Raza, G., Begum, F., Hussain, S.A., Abbas, Q., Ali, M., 2015. Diameter size class distribution of *Pinus gerardiana* Wall. ex D. Don from Gohar Abad Valley district Diamer, Gilgit-Baltistan. *Pakistan. J. Biodivers. Environ. Sci.* 6, 50–56.
- Kolb, T.E., Stone, J.E., 2000. Differences in leaf gas exchange and water relations among species and tree sizes in an Arizona pine-oak forest. *Tree Physiol.* 20 (1), 1–12.
- Kraus, D., Goldammer, J.G., 2007. Fire Regimes and Ecosystems: An Overview of Fire Ecology in Tropical Ecosystems. *Forest fires in India – workshop proceedings*, pp. 9–13.
- Kumar, R., Shamet, G.S., Mehta, H., Alam, N.M., Kaushal, R., Chaturvedi, O.P., Sharma, N., Khaki, B.A., Gupta, D., 2016. Regeneration complexities of *Pinus gerardiana* in dry temperate forests of Indian Himalaya. *Environ. Sci. Pollut. Res.* 23 (8), 7732–7743. <https://doi.org/10.1007/s11356-015-6010-5>.
- Kumar, S., Kumari, R., 2019. Effects of soil on understory species composition in three forests of Kumaun Himalaya. *Indian J. Ecol.* 46, 250–254.
- Kutiel, P., Naveh, Z., 1987. Soil properties beneath *Pinus halepensis* and *Quercus calliprinos* trees on burned and unburned mixed forest on Mt. Carmel, Israel. *For. Ecol. Manage.* 20, 11–24.
- Lal, P., Butola, J.S., Khanduri, V.P., 2017. Community structure of Dandachali Forest of Tehri forest division, North-West Himalaya. *J. Plant Dev. Sci.* 9, 933–939.
- Lawes, M.J., Richards, A., Dathe, J., Midgley, J.J., 2011. Bark thickness determines fire resistance of selected tree species from fire-prone tropical savanna in north Australia. *Plant Ecol.* 212, 2057–2069. <https://doi.org/10.1007/s11258-011-9954-7>.
- Lawes, M.J., Midgley, J.J., Clarke, P.J., 2013. Costs and benefits of relative bark thickness in relation to fire damage: a savanna/forest contrast. *J. Ecol.* 101, 517–524.
- Lee, W.K., Gadow, K.V., Chung, D.J., Lee, J.K., Shin, M.Y., 2004. DBH growth model for *Pinus densiflora* and *Quercus variabilis* mixed forests in central Korea. *Ecol. Modell.* 176, 187–200.
- Loshali, D.C., Singh, R.P., 1992. Partitioning of rainfall by three Central Himalayan forests. *For. Ecol. Manage.* 53, 99–105.
- Majila, B.S., Kala, C.P., 2010. Forest structure and regeneration along the altitudinal gradient in the Binsar Wildlife Sanctuary, Uttarakhand Himalaya, India. *Russ. J. Ecol.* 41, 75–83.
- Matlack, G.R., Gleeson, S.K., Good, R.E., 1993. Treefall in a mixed oak-pine coastal plain forest: immediate and historical causation. *Ecology* 74, 1559–1566.
- Meng, R., Wu, J., Zhao, F., Cook, B.D., Hanavan, R.P., Serbin, S.P., 2018. Measuring short-term post-fire forest recovery across a burn severity gradient in a mixed pine-oak forest using multi-sensor remote sensing techniques. *Remote Sens. Environ.* 210, 282–296.
- Murata, M., Kanetani, S., Nara, K., 2017. Ectomycorrhizal fungal communities in endangered *Pinus amamiana* forests. *PLoS one* 12, e0189957. <https://doi.org/10.1371/journal.pone.0189957>.
- Nagendra, H., Tucker, C., Carlson, L., 2004. Monitoring parks through remote sensing: studies in Nepal and Honduras. *Environ. Manage.* 34, 748–760. <https://doi.org/10.1007/s00267-004-0028-7>.
- Naing, T.Z., 2003. Ecology of the white-browed Nuthatch *Sitta victoriae* in Natmataung National Park, Myanmar, with notes on other significant species. *Forktail* 19, 57–62.
- Nautiyal, A., 2015. Is chir pine displacing banj oak in the Central Himalaya? Socioeconomic implications for local people and the conservation of oak forest biodiversity. University of Arkansas, Fayetteville, p. 385. Ph.D. thesis.
- Pellegrini, A.F.A., Anderegg, W.R.L., Paine, C.E.T., Hoffmann, W.A., Kartzinel, T., Rabin, S.S., et al., 2017. Convergence of bark investment according to fire and climate structures ecosystem vulnerability to future change. *Ecol. Lett.* 20, 307–316. <https://doi.org/10.1111/ele.12725>.
- Park, A.D., 2001. Environmental influences on post-harvest natural regeneration in Mexican pine-oak forests. *For. Ecol. Manage.* 144, 213–228.
- Pausas, J.G., 2015. Bark thickness and fire regime. *Funct. Ecol.* 29, 315–327.
- Perez-Suarez, M., Arredondo-Moreno, J.T., Huber-Sannwald, E., Serna-Perez, A., 2014. Forest structure, species traits and rain characteristics influences on horizontal and vertical rainfall partitioning in a semiarid pine-oak forest from Central Mexico. *Ecology* 7, 532–543.
- Perez-Suarez, M., Arredondo-Moreno, J.T., Huber-Sannwald, E., Vargas-Hernandez, J.J., 2009. Production and quality of senesced and green litterfall in a pine-oak forest in central –northwest Mexico. *For. Ecol. Manage.* 258, 1307–1315.
- Perez-Suarez, M., Arredondo-Moreno, T., Huber-Sannwald, E., 2012. Early stage of single and mixed leaf-litter decomposition in semiarid forest pine-oak: the role of rainfall and microsite. *Biogeochemistry* 108, 245–258.
- Perot, T., Vallet, P., Archaux, F., 2013. Growth compensation in an oak-pine mixed forest following an outbreak of pine sawfly (*Diprion pini*). *For. Ecol. Manage.* 295, 155–161.
- Pimont, F., Dupuy, J.L., Rigolot, E., 2018. A simple model for shrub-strata-fuel dynamics in *Quercus coccifera* L. communities. *Ann. For. Sci.* 75, 44.
- Ralhan, P.K., Saxena, A.K., Singh, J.S., 1982. Analysis of forest vegetation at and around Nainital in Kumaun Himalaya. *Proc. Indian Natl. Sci. Acad.* B48, 121–137.
- Ralhan, P.K., Khanna, R.K., Singh, S.P., Singh, J.S., 1985. Certain phenological characters of the shrub layer of Kumaun Himalayan forests. *Vegetatio* 63, 113–119. <https://doi.org/10.1007/BF00044061>.
- Rana, B.S., Singh, S.P., Singh, R.P., 1989. Biomass and Net Primary Productivity in Central Himalayan Forests along an altitudinal gradient. *For. Ecol. Manage.* 27, 199–218.
- Reyna, T.A., Martinez-Vilalta, J., Retana, J., 2019. Regeneration patterns in Mexican pine-oak forests. *Forest Ecosyst.* 6, 50. <https://doi.org/10.1186/s40663-019-0209-8>.
- Rice, S.K., Westerman, B., Federici, R., 2004. Impacts of the exotic, nitrogen-fixing black locust (*Robinia pseudoacacia*) on nitrogen-cycling in a pine-oak ecosystem. *Plant Ecol.* 174, 97–107.
- Richardson, D.M., 1998. *Ecology and Biogeography of Pinus*. Cambridge University Press.
- Richter, C., Rejmanek, M., Miller, J.E.D., Welch, K.R., Weeks, J.M., Safford, H., 2019. The species diversity x fire severity relationship is hump-shaped in semiarid yellow pine and mixed conifer forests. *Ecosphere* 10 (10), e02882. <https://doi.org/10.1002/ecs2.2882>.
- Rosell, J.A., 2016. Bark Thickness across the angiosperms: more than just fire. *New Phytol.* 211, 90–102. <https://doi.org/10.1111/nph.13889>.
- Saladin, B., 2013. Diversify or specialize? disturbances influence trait evolution in *Pinus*. University of Zurich, Switzerland, p. 88. M.Sc. thesis.
- Sen, A., Quilho, T., Pereira, H., 2011. Bark anatomy of *Quercus cerris* L. var. *cerris* from Turkey. *Turk. J. Botany* 35. <https://doi.org/10.3906/bot-1002-33>.
- Sen, I., Gök, A., 2009. Leaf beetle communities (Coleoptera: Chrysomelidae) of two mixed forest ecosystems dominated by pine-oak-hawthorn in Isparta province. *Ann. Zool. Fennici* 46, 217–232.
- Schwilch, D.W., Brown, T.E., Lackey, R., et al., 2016. Post-fire resprouting oaks (genus: *Quercus*) exhibit plasticity in xylem vulnerability to drought. *Plant Ecol.* 217, 697–710. <https://doi.org/10.1007/s11258-016-0568-y>.
- Shahabuddin, G., Goswami, R., Gupta, M., 2017. An annotated checklist of the birds of banj oak-chir pine forests in Kumaun, Uttarakhand. *Indian Birds* 13, 29–36.
- Shalizi, M.N., Khurram, S., Groninger, J.W., Ruffner, C.M., Burey, O.T., 2018. Indigenous knowledge and stand characteristics of a threatened tree species in a highly insecure area: Chilgoza pine in Afghanistan. *For. Ecol. Manage.* 413, 1–8.
- Sharma, D.P., Singh, M., 2010. Assessing the Carbon Sequestration potential of subtropical pine forest in North-western Himalayas- A GIS approach. *J. Indian Soc. Remote Sens.* 38, 247–253.
- Sheffer, E., 2012. A review of the development of Mediterranean pine-oak ecosystems after land abandonment and afforestation: are they novel ecosystems? *Ann. For. Sci.* 69, 429–443.
- Silva, I.C., Rodriguez, H.G., 2001. Interception loss, throughfall and stemflow chemistry in pine and oak forests in northeastern Mexico. *Tree Physiol.* 21, 1009–1013.
- Singer, A., Ganor, E., Fried, M., Shamay, Y., 1996. Throughfall deposition of Sulfur to a mixed oak and pine forest in Israel. *Atmos. Environ.* 30, 3881–3889.
- Singh, A.K., 2010. Probable Agricultural Biodiversity Heritage Sites in India: VI. The Northeastern Hills of Nagaland, Manipur, Mizoram, and Tripura. *Asian Agri-Hist.* 14, 217–243.
- Singh, J.S., Singh, S.P., 1992. *Forests of Himalaya: Structure, Gyanodaya Prakashan, Nainital, India, Functioning and Impact of Man*, p. 257.
- Singh, R.D., Gumber, S., Tewari, P., Singh, S.P., 2016. Nature of forest fires in Uttarakhand: frequency, size and seasonal patterns in relation to pre-monsoonal environment. *Curr. Sci.* 111, 398–403.
- Singh, S.P., Gumber, S., Singh, R.D., Pandey, R., 2021. Differentiation of diploxylon and haploxylon pines in spatial distribution, and adaptational traits. *Acta Ecol. Sin.* <https://doi.org/10.1016/j.chnaes.2021.07.007>.

- Singh, S.P., Inderjit, Singh, J.S., Majumdar, S., Moyano, J., Nunez, M.A., Richardson, D. M., 2018. Insights on the persistence of pines (*Pinus* species) in the Late Cretaceous and their increasing dominance in the Anthropocene. *Ecol. Evol.* 8, 10345–10359. <https://doi.org/10.1002/ece3.4499>.
- Singh, S.P., Negi, G.C.S., Pant, M.C., Singh, J.S., 1992. Economic considerations of agroecosystem and resultant deforestation in Central Himalaya, in: Agrawal, A. (Eds.), *The price of forests: Proceedings of seminar on Economics of Sustainable use of Forest Resources*, Centre for Science and Environment, New Delhi, pp. 291–296.
- Singh, S.P., Singh, J.S., 1986. Structure and function of the Central Himalayan oak forests. *Proc. Indian Acad. Sci.* 96, 159–189. <https://doi.org/10.1007/BF03053301>.
- Singh, S.P., Zobel, D.B., Garkoti, S.C., Tewari, A., Negi, C.M.S., 2006. Patterns in water relations of central Himalayan trees. *Trop. Ecol.* 47, 159–182.
- Slazak, A., Freese, D., Matos, E.D.S., Huttli, R.F., 2010. Soil organic phosphorus fraction in pine-oak forest stands in Northeastern Germany. *Geoderma* 158, 156–162.
- Strand, E.K., Satterberg, K.L., Hudak, A.T., et al., 2019. Does burn severity affect plant community diversity and composition in mixed conifer forests of the United States Intermountain West one decade post fire? *Fire Ecol.* 15, 25. <https://doi.org/10.1186/s42408-019-0038-8>.
- Subedi, B.P., 1998. Participatory utilization and conservation of Medicinal and Aromatic plants: A case from western Nepal Himalaya. In: *International Conference on Medicinal Plants*, p. 13.
- Sultan, A., Nazir, K., Ayub, N., 2001. Some common species of fleshy macromycetes (Discomycetes, Gasteromycetes and Agaricales) from Margalla Hills National Park and adjacent areas. *Pak. J. Bot.* 33, 709–721.
- Teskey, R.O., Saveyn, A., Steppe, K., McGuire, M.A., 2008. Origin, fate and significance of CO₂ in tree stems. *New Phytol.* 177, 17–32.
- Tewari, J.C., Rikhari, H.C., Singh, S.P., 1989. Compositional and Structural features of certain tree stands along an elevational gradient in central Himalaya. *Vegetatio* 85, 117–120. <http://www.jstor.org/stable/20038560>.
- Thomas, R.E., Bennett, N.D., 2014. Estimating bark thickness of common Appalachian hardwoods, in: Groninger, W.J., Holzmueller, J.E., Nielsen, K.C., Daniel, C.D. (Eds.) *Proc. of 19th Central Hardwood Forest Conference*; Carbondale, IL. General Technical Report NRS-P-142, Newtown Square, PA: U.S. Department of Agriculture, Forest Service, Northern Research Station, pp 283–294.
- Troup, R.S., 1921. *The Silviculture of Indian Trees Volume III*, 1186.
- Turkmen, N., Duzenli, A., 2011. Early post-fire changes in *Pinus brutia* forests (Amanos Mountain, Turkey). *Acta. Bot. Croat.* 70, 9–21.
- Tyree, M.T., Sperry, J.S., 1988. Do woody plants operate near the point of catastrophic xylem dysfunction caused by dynamic water stress? *Plant Physiol.* 88, 574–580.
- Wakeel, A., Rao, K.S., Maikhuri, R.K., Saxena, K.G., 2005. Forest Management and land use/ cover changes in a typical micro watershed in the mid elevation zone of central Himalaya. India. *For. Ecol. Manage.* 213, 229–242.
- Wozniwoda, B., Dyderski, M.K., Kobus, S., Parzych, A., Jagodzinski, A.M., 2019. Natural regeneration and recruitment of native *Quercus robur* and introduced *Q. rubra* in European oak-pine mixed forests. *For. Ecol. Env.* 449, 117473 <https://doi.org/10.1016/j.foreco.2019.117473>.
- Zavala, M.A., Zea, E., 2004. Mechanisms maintaining biodiversity in Mediterranean pine-oak forests: insights from a spatial simulation model. *Plant Ecol.* 171, 197–207.
- Zavala, M.A., Galindo-Jaimes, L., Gonzalez-Espinosa, M., 2010. O Models of regional and local stand composition and dynamics of pine-oak forests in the central highlands of Chiapas (Mexico): Theoretical and Management Implications. In: Newton, A.C. (Ed.), *Biodiversity Loss and Conservation in Fragmented Forest Landscapes: The Forests of Montane Mexico and Temperate South America*. CAB International, pp. 223–243.
- Zhang, G., Hui, G., Zhang, G., Zhao, Z., Hu, Y., 2019. Telescope method for characterizing the spatial structure of a pine-oak mixed forest in the Xiaolong Mountains. China. *Scand. J. For. Res.* 34, 751–762. <https://doi.org/10.1080/02827581.2019.1680729>.